Digital Logic Circuits (Spring 2026)

HW #2 (Due: April 8, 2026)

11. Apply DeMorgan's theorems to the following:

(a) $\overline{\overline{(\overline{ABC})(\overline{EFG})} + \overline{(\overline{HIJ})(\overline{KLM})}}$ **(b)** $\overline{(A + \overline{B\overline{C}} + CD)} + \overline{\overline{BC}}$

(c) $\overline{\overline{\overline{(A + B)}\,\overline{(C + D)}\,\overline{(E + F)}\,\overline{(G + H)}}}$

15. Draw the logic circuit represented by each expression:

(a) $AB + \overline{AB}$ **(b)** $ABCD$

(c) $A + BC$ **(d)** $ABC + D$

21. Using Boolean algebra, simplify the following expressions:

(a) $CE + C(E + F) + \overline{E}(E + G)$ **(b)** $\overline{B}\,\overline{C}D + \overline{(B + C + D)} + \overline{B}\,\overline{C}\,\overline{D}E$

(c) $(C + CD)(C + \overline{C}D)(C + E)$ **(d)** $BCDE + BC(\overline{DE}) + (\overline{BC})DE$

(e) $BCD[BC + \overline{D}(CD + BD)]$

33. Develop a truth table for each of the SOP expressions:

(a) $\overline{A}B + AB\overline{C} + \overline{A}\,\overline{C} + A\overline{B}C$ **(b)** $\overline{X} + Y\overline{Z} + WZ + X\overline{Y}Z$

42. Expand each expression to a standard SOP form:

(a) $AB + A\overline{B}C + ABC$ **(b)** $A + BC$

(c) $A\overline{B}\,\overline{C}D + AC\overline{D} + B\overline{C}D + \overline{A}BC\overline{D}$ **(d)** $A\overline{B} + A\overline{B}\,\overline{C}D + CD + B\overline{C}D + ABCD$

43. Minimize each expression in Problem 42 with a Karnaugh map.

49. Use a Karnaugh map to simplify each expression to minimum POS form:

(a) $(A + \overline{B} + C + \overline{D})(\overline{A} + B + \overline{C} + D)(\overline{A} + \overline{B} + \overline{C} + \overline{D})$

(b) $(X + \overline{Y})(W + \overline{Z})(\overline{X} + \overline{Y} + \overline{Z})(W + X + Y + Z)$

11. Apply DeMorgan's theorems to the following:

(a) $\overline{\overline{\overline{(ABC)}\,\overline{(EFG)}}} + \overline{\overline{\overline{(HIJ)}\,\overline{(KLM)}}}$ (b) $\overline{(A + \overline{B\overline{C}} + CD)} + \overline{\overline{BC}}$

(c) $\overline{\overline{\overline{(A + B)}\,\overline{(C + D)}\,\overline{(E + F)}\,\overline{(G + H)}}}$

$\overline{B\overline{C}} = \overline{B} + C$

(a) $= \overline{\overline{\overline{(ABC)}\,\overline{(EFG)}}}\ \overline{\overline{\overline{(HIJ)}\,\overline{(KLM)}}}$

$= \overline{(ABC)}\,\overline{(EFG)}\,\overline{(HIJ)}\,\overline{(KLM)}$

$= (\overline{A} + \overline{BC}) \cdots$

$= (\overline{A} + \overline{B} + \overline{C}) \cdots$

$= (\overline{A} + \overline{B} + \overline{C})(\overline{E} + \overline{F} + \overline{G})(\overline{H} + \overline{I} + \overline{J})(\overline{K} + \overline{L} + \overline{M})$

(b) $= \overline{(A + \overline{B} + C + CD)} + BC$

$= \overline{(A + \overline{B} + C)} + BC$

$= B(\overline{A + C})\,BC$

$= BC(\overline{A + C})$

$= BC\overline{A}\,\overline{C} = 0$

(c) $= \overline{(A+B)}\,\overline{(C+D)}\,\overline{(E+F)}\,\overline{(G+H)}$

$= \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}\,\overline{E}\,\overline{F}\,\overline{G}\,\overline{H}$

42. Expand each expression to a standard SOP form:

(a) $AB + A\overline{B}C + ABC$ (b) $A + BC$

(c) $A\overline{B}\,\overline{C}D + AC\overline{D} + B\overline{C}D + \overline{A}BC\overline{D}$ (d) $A\overline{B} + A\overline{B}\,\overline{C}D + CD + B\overline{C}D + ABCD$

(a) $AB(C + \overline{C}) + A\overline{B}C + ABC$

$= ABC + A\overline{B}C + AB\overline{C}$

(b) $A(B + \overline{B})(C + \overline{C}) + (A + \overline{A})BC$

$= ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C} + ABC + \overline{A}BC$

$= ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}BC$

(c) $A\overline{B}\,\overline{C}D + A(B + \overline{B})C\overline{D} + (A + \overline{A})B\overline{C}D + \overline{A}BC\overline{D}$

$= A\overline{B}\,\overline{C}D + ABC\overline{D} + A\overline{B}C\overline{D} + AB\overline{C}D + \overline{A}B\overline{C}D + \overline{A}BC\overline{D}$

(d) $= A\overline{B}(C + \overline{C})(D + \overline{D}) + A\overline{B}\,\overline{C}D + (A + \overline{A})(B + \overline{B})CD$

$+ (A + \overline{A})B\overline{C}D + ABCD$

$= A\overline{B}CD + A\overline{B}C\overline{D} + A\overline{B}\,\overline{C}D + A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}D + ABCD + \overline{A}BCD$

$+ \overline{A}BCD + \overline{A}\,\overline{B}CD + AB\overline{C}D + \overline{A}B\overline{C}D + ABCD$

$= ABCD + A\overline{B}CD + \overline{A}BCD + A\overline{B}\,\overline{C}D + A\overline{B}C\overline{D} + A\overline{B}\,\overline{C}\,\overline{D}$

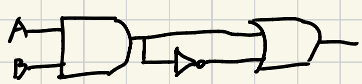
$+ \overline{A}\,\overline{B}CD + AB\overline{C}D + \overline{A}B\overline{C}D$

15. Draw the logic circuit represented by each expression:

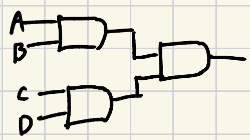
(a) $AB + \overline{AB}$ (b) $ABCD$

(c) $A + BC$ (d) $ABC + D$

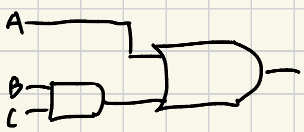
(a)



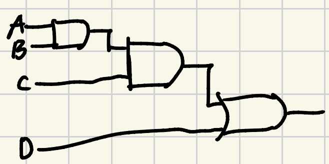
(b)



(c)



(d)



21. Using Boolean algebra, simplify the following expressions:

(a) $CE + C(E + F) + \overline{E}(E + G)$ (b) $\overline{B}\,\overline{C}D + \overline{(B + C + D)} + \overline{B}\,\overline{C}\,\overline{D}E$

(c) $(C + CD)(C + \overline{C}D)(C + E)$ (d) $BCDE + BC(\overline{DE}) + (\overline{BC})DE$

(e) $BCD[BC + \overline{D}(CD + BD)]$

(a) $CE + CE + CF + 0 + \overline{E}G$

$= CE + CF + \overline{E}G$

$= C(E + F) + \overline{E}G$

(b) $\overline{B}\,\overline{C}D + \overline{B}\,\overline{C}\,\overline{D} + \overline{B}\,\overline{C}\,\overline{D}E$

$= \overline{B}\,\overline{C}D + \overline{B}\,\overline{C}\,\overline{D}$

$= \overline{B}\,\overline{C}$

(c) $C(C + D)(C + E)$

$= C$

(d) $BC + \overline{BC}DE = BC + DE$

(e) $BCD + 0 = BCD$

43. Minimize each expression in Problem 42 with a Karnaugh map.

(a)

AB \ C	0	1
00	0	0
01	0	0
11	1	1
10	0	1

$AB + AC$

$= A(B + C)$

(b)

AB \ C	0	1
00	0	0
01	0	1
11	1	1
10	1	1

$BC + AB + A\overline{C} + A\overline{B}$

$= A + BC$

33. Develop a truth table for each of the SOP expressions:

(a) $\overline{A}B + AB\overline{C} + \overline{A}\,\overline{C} + A\overline{B}C$ (b) $\overline{X} + Y\overline{Z} + WZ + X\overline{Y}Z$

(a) $\overline{A}B = \overline{A}B(C + \overline{C})$

$= \overline{A}BC + \overline{A}B\overline{C}$

$\overline{A}\,\overline{C} = \overline{A}\,\overline{C}(B + \overline{B})$

$= \overline{A}B\overline{C} + \overline{A}\,\overline{B}\,\overline{C}$

$\overline{A}BC + \overline{A}B\overline{C} + AB\overline{C} + \overline{A}B\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C$

$= \overline{A}BC + \overline{A}B\overline{C} + AB\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C$

A	B	C	output
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(b) $\bar{X} \Rightarrow X=0$

$Y\bar{Z} \Rightarrow Y=1, \bar{Z}=0$

$WZ \Rightarrow W=1, Z=1$

$X\bar{Y}Z \Rightarrow X=1, Y=0, Z=1$

W	X	Y	Z	output
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

43 (c)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$B\bar{C}D + A\bar{C}D + BC\bar{D} + AC\bar{D}$

$\bar{C}D(A+B) + (A+B)C\bar{D}$

$= (A+B)(\bar{C}D + C\bar{D})$

43 (d)

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	0	1	1	0
10	1	1	1	1

$CD + BD + A\bar{B}$

$= D(B+C) + A\bar{B}$

49. Use a Karnaugh map to simplify each expression to minimum POS form:

(a) $(A + \bar{B} + C + \bar{D})(\bar{A} + B + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})$

(b) $(X + \bar{Y})(W + \bar{Z})(\bar{X} + \bar{Y} + \bar{Z})(W + X + Y + Z)$

(a)
① A=0 B=1 C=0 D=1
② A=1 B=0 C=1 D=0
③ A=1 B=1 C=1 D=1

cannot simplify

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	1
11	1	1	0	1
10	1	1	1	0

(b)
① WXYZ = X01X
② WXYZ = 0XX1
③ X111
④ 0000

W+X \ Y+Z	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	1	0	1
10	1	1	0	0

$(W+X)(\bar{Y}+\bar{Z})(W+\bar{Z})(X+\bar{Y})$